

Trevion Duncan

ITAI 2372 — Artificial Intelligence Portfolio

- Explored how Artificial Intelligence is being applied across real-world industries.
- Studied emerging AI trends, industry use cases, ethical concerns, and future applications.
- Coursework covered financial services, agriculture, entertainment & media, and retail/supply chain.

Highlight #1 — AI in Fraud Detection

Financial Services Case Study

- Analyzed how banks use AI and machine learning to identify unusual transaction activity.
- Explored transaction monitoring, pattern recognition, and real-time risk scoring.
- Key learning: AI can analyze large amounts of transaction data quickly and identify patterns humans may miss.
- Challenges: false positives, data quality, privacy, bias, and implementation costs.

Highlights #2–3 — AI Across Industries

Agriculture • Entertainment • Supply Chain

- Agriculture: precision farming, drones, sensors, satellite crop monitoring, and automated irrigation.
- Entertainment: AI-generated music, production assistance, recommendations, copyright, and artist rights.
- Retail/Supply Chain: demand forecasting, inventory optimization, delivery routing, and disruption prediction.
- Key learning: AI can support prediction, automation, efficiency, and decision-making across industries.

Real-World Applications

Turning AI Knowledge Into Business Value

- Financial Services — support fraud detection and risk analysis.
- Agriculture — improve crop monitoring, resource use, and predictive decisions.
- Entertainment & Media — support creative workflows, production, and recommendations.
- Retail & Supply Chain — improve forecasting, inventory, routing, and operational efficiency.
- Responsible AI — consider privacy, fairness, transparency, cybersecurity, and human oversight.